Can primordial deuterium fix the dark-matter density? DESI DR2 distances read with Ω_c/Ω_b = 3√3 and no CMB, the Hubble constant, $a_o = cH_o/6$, and a registered test for DESI beyond DR2

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Abstract

Standard big-bang nucleosynthesis, used here as a dictionary of laboratory data, constrains the baryon density ω_b with the deuterium measured in quasar absorbers (and helium, in the primary dictionary). If the dark-matter density is ω_c = 3√3·ω_b, the 13 DESI DR2 BAO distances are read with one cosmological parameter fewer than in flat ΛCDM. No CMB data enter the fits. Results. A premise of the framework on time treats the standard account of the first minutes as a reading in the present. Six translations into measured quantities were registered before computing, and run with PRyMordial (NACRE II rates with LUNA): The predicted lithium stays about 3.5 times the observed value in every reading that holds. Registered test , before the data: the first DESI release beyond DR2, read with the same chain without the CMB. $\Delta\chi^2 \le 4$ «holds», 4-9 «tension», > 9 «the reading 3√3 does not read BAO + BBN». A secondary block compares $a_0 = cH_0/6$ with the first g† measured with a total error $\le 0.06 \cdot 10^{-10}$. An addendum registered before the data reports the same computation with evolving dark energy next to the main block, with a qualifier fixed in advance. Decision programs with tests are included.

Can the deuterium of the first minutes fix how much dark matter there is? The DESI distances read with the 3√3 ratio and without the background radiation, the Hubble constant, $a_o = cH_o/6$ and a registered test for the next DESI data

Student version of MM record 2.58 (DOI 10.5281/zenodo.23521754), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.58 of the Mathematical Materiality (MM) series, October 2026. It continues MM 2.54 (DOI 10.5281/zenodo.23418156). There the ratio between dark matter and ordinary matter, Ω_c/Ω_b = 3√3, was registered. The data used included the cosmic microwave background (CMB). This work then feeds the chain of MM 2.56 (DOI 10.5281/zenodo.23506401) on wide binaries.

Before starting: the words you need

The first minutes of the universe. In the first minutes the universe was extremely hot. Protons and neutrons joined together and formed the first light nuclei. This process is called *big-bang nucleosynthesis*, BBN for short.

- **Deuterium** (D) is hydrogen with one more neutron in the nucleus.
- **Helium-4** (^{4}He) has two protons and two neutrons.
- **Lithium** (Li) is an even heavier nucleus, produced in very small amounts.

How much deuterium formed depends on how much ordinary matter there was. For this reason the deuterium measured today says how much ordinary matter the universe has.

ω_b and ω_c. ω_b measures how much ordinary matter there is in the universe. Ordinary matter is made of baryons. ω_c measures how much cold dark matter there is. Ω_c/Ω_b is their ratio.

The ratio 3√3. In this work it is imposed that dark matter is 3√3 times ordinary matter. As a formula: $\omega_c = 3\sqrt{3}\cdot\omega_b = 5.196\cdot\omega_b$. In standard cosmology, ΛCDM, ω_c is instead free.

The BAO and the sound horizon. In the young universe matter vibrated like a gas crossed by sound waves. These waves left a typical distance in the distribution of galaxies: the *baryon acoustic oscillations* (BAO). The distance that the waves travelled up to the *drag epoch*, when ordinary matter separated from light, is called the *sound horizon*, r_d. It acts as a cosmic ruler. DESI is a large survey of galaxies that measures these distances.

The Hubble constant, H_0. It says how fast the universe is expanding today. It is measured in km s^{-1} Mpc^{-1}: kilometres per second for each megaparsec of distance. The measurements do not agree with each other. From the background radiation (Planck) comes 67.36. From nearby stars (Riess and colleagues) comes 73.04.

a_0. It is the acceleration scale of galaxies, in the *radial acceleration relation*. That relation links the acceleration observed in galaxies to the acceleration of the visible matter alone. Here we read $a_0 = cH_0/6$, where c is the speed of light.

The χ^2, the σ and the two z. The χ^2 (read *chi-squared*) is an error score. The lower it is, the better a model follows the data. $\Delta\chi^2$ is the difference between two models. σ is the typical error. Careful: in this text the letter z has two meanings.

- As *redshift*, z says how far away a galaxy is. The larger z is, the farther away and the further back in time we look.
- As a *z score*, it says by how many σ a value is distant from another. For example z = −0.38 means 0.38 typical errors below.

The dictionary. BBN is used here as a *dictionary*. It translates the numbers measured in the laboratory into cosmological quantities. The laboratory numbers are the nuclear reactions, the neutron lifetime and others.

The prior. It is a piece of information given to a calculation at the start. For example: *ω_b is about 0.02218, with a certain error*.

In short

The chain in five steps.

- Standard nucleosynthesis, with the measured deuterium (and helium, in the primary dictionary), constrains ω_b.
- $\omega_c = 3\sqrt{3}\cdot\omega_b$ is imposed. Then the 13 BAO distances of DESI DR2 are read with one cosmological parameter fewer than in the flat ΛCDM model.
- The fit costs only $\Delta\chi^2 = 0.09$. It gives $H_0 = 68.52 \pm 0.59$ km s^{-1} Mpc^{-1}, the value already known from BAO + BBN.
- Then $a_0 = cH_0/6 = 1.110\cdot10^{-10}$ m s^{-2}. It is at −0.4σ from the value of SPARC, a collection of nearby galaxies.
- **The other way round:** the BAO alone, with 3√3, predict $D/H = (2.40 \pm 0.31)\cdot10^{-5}$. The measured value is 2.51. It is a predictive check, but not independent of the ratio test.

A test on the first DESI release after DR2 is registered here, before those data come out.

What it is, and what it is not. It is a consistency check of one reading: 3√3, chosen knowing the data. It is not a derivation. Its outcome concerns that reading, not the MM framework.

1. The chain

1. **Deuterium → ω_b.** Standard nucleosynthesis (BBN) acts as a dictionary. It is a function of laboratory data. First of all the nuclear *cross sections*, that is, how easy it is for two nuclei to react. Among these is the reaction $d(p,\gamma)^3He$ measured by LUNA. Then the neutron lifetime, G, N_eff and T_γ. The primary dictionary uses D and ^4He. A dictionary with deuterium alone (D-4) serves as a check.
2. **× 3√3 → ω_c.** $\omega_c = 3\sqrt{3}\cdot\omega_b = 5.196\cdot\omega_b$. In ΛCDM ω_c would be free.
3. **→ the sound horizon r_d and H_0.** With ω_b and ω_c fixed, r_d is fixed (the CLASS program computes it).

The DESI DR2 BAO then leave free a single cosmological parameter, H_0. ω_b remains a fitted parameter, constrained by the BBN prior.

4. **→ $a_0 = cH_0/6$ → galaxies (SPARC) and wide binaries (the chain of MM 2.56).**

No CMB data enter the fits. Only one Level 2 check (T4, section 5) compares the result with Planck, afterwards.

2. The data

These are the values read on the sources.

quantity	source	value
BAO distances	DESI DR2 (DESI Collaboration 2025, PRD 112, 083515, arXiv:2503.14738), public Gaussian likelihood files (CobayaSampler/bao_data, commit b7b8a36e; not included, downloaded by scarica_dati.py with SHA-256 check)	13 measurements DV/r_d, DM/r_d, DH/r_d, z = 0.295-2.33, with covariance
primordial D/H	Cooke, Pettini & Steidel 2018 (ApJ 855, 102), eq. 10; PDG 2026 review 24, eq. 24.2 (12 systems)	10^5 D/H = 2.527 ± 0.030; 10^6 D/H = 25.08 ± 0.29
ω_b from BBN (dictionary D-1, primary)	Schöneberg 2024 (arXiv:2401.15054), eq. 5.1: PRyMordial NACRE II with LUNA, marginalised over nuclear rates, PDG 2023 abundances of D **and** ^{4}He; the prior used by DESI DR2 (eq. 14)	0.02218 ± 0.00055
Y_p	PDG 2026 eq. 24.3; Aver et al. 2022 (MNRAS 510, 373) eq. 6	0.245 ± 0.003; 0.2448 ± 0.0033
Li/H observed; predicted	PDG 2026 eq. 24.4; eq. 24.10	$(1.45 \pm 0.25)\cdot10^{-10}$ (strictly a lower bound); $(4.72 \pm 0.7)\cdot10^{-10}$
neutron lifetime	PDG 2026 listings; UCNτ 2025 (PRC 111, 045501); Yue et al. 2013 (PRL 111, 222501); J-PARC 2024 (arXiv:2412.19519)	878.3 ± 0.4 s; 877.82 ± 0.22 +0.20/−0.17 s; 887.7 ± 2.3 s; 877.2 ± 1.7 +4.0/−3.6 s
H_0 for comparison	Planck 2018 VI eq. 14; Riess et al. 2022; Breuval et al. 2024	67.36 ± 0.54; 73.04 ± 1.04; 73.17 ± 0.86
a_0 of SPARC	McGaugh, Lelli & Schombert 2016 (PRL 117, 201101)	$g\dagger = 1.20 \pm 0.02$ (random) ± 0.24 (systematic) $\cdot10^{-10}$ m s^{-2}
BBN tables	PRyMordial (Burns, Tait & Valli 2024, EPJC 84, 86), commit 725d8a8d, file PRyM_Yp_DH_cosmoMC_2023.dat (GPL v3; not included, downloaded by scarica_dati.py)	Y_p and D/H with theoretical errors as functions of ω_b and ΔN_eff

How to read some rows.

- **D/H** is the number of deuterium nuclei for each hydrogen nucleus. 10^5 D/H = 2.527 means that there are 2.527 deuterium nuclei for every hundred thousand hydrogen nuclei. It is measured in the light of distant gas clouds poor in stars (12 systems).
- **Y_p** is the mass fraction of helium-4 produced in the first minutes: about 0.245.
- **The neutron lifetime** is measured in two ways. In the *bottle* method the neutrons are shut in a container and one counts how many remain. In the *beam* method a beam of neutrons in flight is observed. The two methods give different values: for example 878.3 s with the bottle and 887.7 s with the beam.
- **H_0 for comparison:** Planck gives 67.36; Riess and colleagues 73.04; Breuval and colleagues 73.17.

3. The method (Level 1)

The model.

- Flat ΛCDM, with N_eff = 3.044: there is one neutrino with a mass of 0.06 eV. The temperature of the background radiation is T_γ = 2.7255 K.
- The sound horizon r_d is computed by the CLASS 3.4.1 program on a grid of values of (ω_b, ω_c). Between the grid points it is interpolated. The distances are computed with an integrator of our own.

Two fits. The score is $\chi^2 = \chi^2_BAO + ((\omega_b - \omega_b,D)/\sigma_D)^2$: 14 terms in all.

- **ΛCDM:** ω_b, ω_c and h free (11 degrees of freedom).
- **MM:** ω_c = 3√3·ω_b, with ω_b and h free (12 degrees of freedom).

The priors.

- Flat: h ∈ [0.55, 0.85], ω_c ∈ [0.085, 0.155].
- ω_b ∈ [0.0196, 0.0249]. It is the edge of the r_d grid, at least 4.7σ from the centre of D-1.
- The distributions of the possible values come from a Metropolis sampler with 4 chains. The convergence checks are good ($\hat{R} \leq 1.0007$ also on the split chains, ESS ≥ 11,800).

The dictionaries, all fixed in advance.

- D-1, primary.
- D-2, the same with Cooke's D/H.
- D-3, PDG 2026.
- D-4, deuterium only: the PDG 2026 D/H through the PRyMordial table, with the ln V term of the Gaussian likelihood.
- D-5, Cooke 2018 before LUNA (historical).

The validations, all passed.

- The interpolation of r_d agrees with CLASS within 0.00003 Mpc, and the distances within $1.3 \cdot 10^{-6}$.
- The BAO alone give Ω_m = 0.2975 ± 0.0087 and h·r_d = 101.53 ± 0.74. DESI, in its equation 17, gives 0.2975 ± 0.0086 and 101.54 ± 0.73.
- ΛCDM + BBN gives H_0 = 68.55 ± 0.60. DESI, in its equation 19, gives 68.51 ± 0.58.

4. The results

quantity	value
χ^2_min: ΛCDM; MM	10.282; 10.369
$\Delta\chi^2$ (one parameter fewer)	**0.087** (registered zone: compatible within 1σ); ΔAIC −1.91, ΔBIC −2.55 (N = 14 by the registered convention)
Ω_c/Ω_b left free (ΛCDM)	5.282 ± 0.228, +0.38σ from 3√3
H_0, MM	**68.52 ± 0.59**; profile $\Delta\chi^2$ = 1: [67.95, 69.05]
H_0, ΛCDM	68.56 ± 0.60
$a_0 = cH_0/6$	**$(1.110 \pm 0.010) \cdot 10^{-10}$ m s^{-2}**; against SPARC: z = −0.38 (with H_0 = 73.04: z = −0.07)
Y_p at ω_b,MM (convention Y_P = 4·Y_He of PRyMordial)	0.2468, +0.6σ from PDG and Aver (not independent with D-1)
DESI subsets (MM)	z < 1.1: 68.57 ± 0.64; z > 1.1: 68.32 ± 0.93
deuterium only (D-4)	$\Delta\chi^2$ 0.023; H_0 = 68.72 ± 0.60; Y_p = 0.2469 (+0.6σ, now an independent check)
sensitivity D-1…D-5	$\Delta\chi^2$ 0.02-0.42; H_0 67.99-68.72

How to read it.

- **The cost of the ratio.** With the ratio 3√3 the score rises only from 10.282 to 10.369: $\Delta\chi^2 = 0.087$. The registered zone for this value is *compatible within 1σ*. There are also the ΔAIC and ΔBIC criteria, which reward models with fewer parameters. They are negative: they slightly favour the model with the fixed ratio.
- **The ratio left free.** In ΛCDM it comes out at 5.282 ± 0.228: at +0.38σ from 3√3.

- **H_0.** With the fixed ratio it comes out at 68.52 ± 0.59. In ΛCDM it comes out at 68.56 ± 0.60: almost the same.
- **a_0.** It comes out at $(1.110 \pm 0.010)\cdot 10^{-10}$ m s^{-2}. Compared with SPARC, z = −0.38. With $H_0 = 73.04$ it would be z = −0.07.
- **Helium.** Y_p comes out at 0.2468, at +0.6σ from the measurements. With the D-1 dictionary it is not an independent check, because D-1 already uses helium.
- **Deuterium only (D-4).** $\Delta\chi^2 = 0.023$, $H_0 = 68.72 \pm 0.60$, Y_p = 0.2469. Here helium becomes an independent check, and it works out (+0.6σ).
- **The sensitivity.** Changing dictionary (from D-1 to D-5), $\Delta\chi^2$ goes from 0.02 to 0.42 and H_0 from 67.99 to 68.72.

The inverse reading.

- In ΛCDM the BAO alone cannot fix ω_b. With $\omega_c = 3\sqrt{3}\cdot\omega_b$, instead, they can.
- The result is ω_b = 0.02316 ± 0.00178 and $H_0 = 69.4 \pm 1.7$, without any information from BBN.
- Through the standard dictionary this predicts **$D/H = (2.40 \pm 0.31)\cdot 10^{-5}$**.
- The measured value is 2.508 ± 0.029: z = −0.34.
- D/H does not enter the BAO fit. But the check uses the same BAO and the same constraint as the Ω_c/Ω_b test. So it is not independent of it.

What the numbers mean.

- The chain is consistent: the BAO are read as well as in ΛCDM, with one parameter fewer.
- H_0 is the BAO + BBN value already known: it is compatibility, not a new number.
- Today the constraint does not narrow H_0 (0.59 against 0.60), because the data already sit close to the ratio.
- $a_0 = cH_0/6$ is consistent with SPARC, but not decisive. The systematic error of SPARC, 20 %, also allows $H_0 = 73$.
- The wide-binary chain of MM 2.56 uses $H_0 = 67.4$. The difference of 1.1 km s^{-1} Mpc^{-1} lies inside the sensitivity range declared there (67.4-73.04). The protocol of 2.56 does not change.

5. Readings of a premise on time (Level 2)

The premise. A premise of the MM framework treats the standard account of the first minutes as a reading made in the present. In this reading there is a single common time for all processes, and no objective beginning.

The translations. The premise was translated in six ways, into quantities that can be measured. The translations were registered before the calculations. The PRyMordial program was run with the NACRE II rates and with LUNA. It used the full network of 63 nuclear reactions.

translation	content	D/H	Y_p	Li/H	outcome
T0 common clock	rates of all processes in one time unit: abundances are ratios of counts, the dictionary is unchanged	$2.525\cdot 10^{-5}$	0.2468 (+0.6σ)	$5.11\cdot 10^{-10}$ = 3.5 × the observed value	D and He read; Li not read
T1 same everywhere	no significant variation of D/H among 12 absorbers (directions, z_abs 2.4-3.6, metallicities), at the sensitivity of these tests	constancy p = 0.30; slope in z_abs p = 0.61; in metallicity p = 0.17; dipole p = 0.64	—	—	compatible (the dipole is weakly informative: 9 of 12 lines of sight lie between 9h and 16h)
T2a expansion rate = local rate (S = 1)	ΔN_eff free	D + He: ΔN = −0.12 ± 0.21, i.e. $S^2 = 0.981 \pm 0.034$	with BAO and 3√3: ΔN = −0.10 ± 0.18	3.55 ×	compatible with S = 1
T2b hidden mirror sector at the same temperature, with gravitating radiation	ΔN_eq = 6.14 before e^+e^- annihilation, 7.44 after	$4.6\text{-}5.0\cdot 10^{-5}$ (+10 to +11σ)	0.306-0.315 (+20 to +23σ)	$2.8\text{-}3.1\cdot 10^{-10}$	**does not hold** for the two equivalent cases examined (the full thermal history of the mirror sector is not simulated); consistent with the assumption of MM 2.44 that hidden radiation does not gravitate
T3 neutron lifetime	bottle 878.3 / beam 888.0 / J-PARC 877.2 s	2.525 / 2.537 / $2.524\cdot 10^{-5}$	+0.59 / +1.24 /	3.5 ×	weak hint for the bottle value ($\Delta\chi^2 = 1.2$); He prefers τ_n = 870 ± 15 s

			+0.51σ		
T4 one present baryon-to-photon ratio	ω_b from D/H alone against the CMB	0.02242 ± 0.00056 against 0.02230 ± 0.00021 (z = +0.20)	0.2468	3.5 ×	compatible with the CMB value

What each row says.

- **T0, a common clock.** The rates of all processes are counted in a single unit of time. The abundances are ratios of counts, and the dictionary does not change. Deuterium and helium are read well. Lithium is not.
- **T1, the same everywhere.** Deuterium is measured in 12 distant gas clouds. The clouds lie in different directions and at different distances (z_abs from 2.4 to 3.6). They also have different amounts of heavy elements. There is no significant variation, at the sensitivity of these tests. The check on the *dipole* (a difference between two sides of the sky) says little: 9 lines of sight out of 12 lie in the same area of the sky, between 9h and 16h.
- **T2a, expansion rate equal to the local rate.** The extra radiation, ΔN_eff, is left free. It comes out compatible with S = 1, that is, with no difference.
- **T2b, a hidden mirror sector at the same temperature, with gravitating radiation.** It is a copy world of ours, with its own light, which weighs like ours. It would add a lot of radiation. Before the e^+e^- annihilation one would have ΔN_eq = 6.14, after it 7.44. The annihilation is the moment when electrons and positrons turned into light. Deuterium would come out at +10/+11σ and helium at +20/+23σ. **This reading does not hold**, in the two cases examined. The full thermal history of the mirror sector is not simulated. The outcome agrees with the assumption of MM 2.44: hidden radiation does not gravitate.
- **T3, the neutron lifetime.** Three values are tried: bottle (878.3 s), beam (888.0 s) and J-PARC (877.2 s). Deuterium changes little. There is a weak hint in favour of the bottle value ($\Delta\chi^2 = 1.2$). Helium prefers τ_n = 870 ± 15 s.
- **T4, a single baryon-to-photon ratio today.** ω_b is obtained from deuterium alone: 0.02242 ± 0.00056. It is compared with the CMB value: 0.02230 ± 0.00021. They are compatible: z = +0.20.

Lithium.

- In every reading that holds, the predicted lithium is about 3.5 times the measured one. Lithium is measured in the old stars of the halo of the Galaxy, poor in metals.
- The difference is 4.3-4.6σ, where the log computes z, with a theoretical error of 15 %.
- No translation was built on lithium, and none reads it.
- It remains a measured fact to look at. The PDG treats the value measured in stars, strictly, as a lower bound.

What changes for the chain: nothing. The translations that hold leave the dictionary of Level 1 as it is. T2b does not hold in the cases examined. T3 leaves the neutron lifetime open, with a weak hint for the value already used.

5b. The same chain with a dark energy that changes over time

The check. It was done on 20-21 October 2026.

- A dark energy that can change over time is used. It is the w_0w_a (CPL) model, flat, with the DESI DR2 priors.
- The data are DESI DR2 + BBN, without and with four supernova samples: Pantheon+, Union3, DES-SN5YR, DES Dovekie.
- The code reproduces the $\Delta\chi^2$ between w_0w_a and ΛCDM that DESI obtains without the CMB: −4.90, −10.07, −13.57, against DESI's −4.9, −10.1, −13.6.

combination	q free in w_0w_a	$\Delta\chi^2$_q in w_0w_a	$\Delta\chi^2$_q in ΛCDM	H_0 with $3\sqrt{3}$ in w_0w_a
DESI+BBN	5.68 ± 0.48	1.85	0.09	64.7 ± 2.2
+ Union3	5.57 ± 0.51	0.51	0.95	65.8 ± 1.0
+ Pantheon+	5.17 ± 0.56	0.02	1.15	67.1 ± 0.9
+ DES-SN5YR	5.49 ± 0.52	0.31	3.09	66.4 ± 0.8
+ DES Dovekie	5.43 ± 0.54	0.17	1.72	67.0 ± 0.8

Here q is the ratio Ω_c/Ω_b.

- **3√3 is compatible in all five combinations**, according to the registered criterion $\Delta\chi^2_q \leq 4$.
- In w_0w_a, however, the error on q doubles. So the check is less stringent.
- In ΛCDM the supernovae push q upwards: the cost of 3√3 rises up to 3.09 with DES-SN5YR. The chain of sections 1-4 does not use supernovae.
- There is a delicate point near the edge $w_0 + w_a \to 0$. There our own sound-horizon correction differs from CLASS by up to 0.9 %. There the calculations were redone with CLASS. Into that region fall 0.00-0.22 % of the samples, and the means stay the same within 0.06σ.

6. The registered prediction

The time limit for the targets is 20 October 2026, at 18:56:04 UTC. The prediction is written before the data.

Block D (main): DESI beyond DR2 + BBN, without CMB. This combination is not covered by the prediction of MM 2.54, which uses CMB + BAO.

- **The target.** The first paper of the DESI collaboration on arXiv after the time limit. It must have BAO beyond DR2 and public Gaussian likelihood files. A check on arXiv just before registration found none.
- **Data and chain.** The same chain, frozen; dictionary D-1.
- **Validity.** W1: the BAO-only Ω_m and h·r_d of the paper. W2: its DESI+BBN H_0. The outcome is suspended in three cases: at the edges of the priors, with chains that do not converge, with a negative $\Delta\chi^2$.
- **The rule.** $\Delta\chi^2 \leq 4$: "the chain holds". From 4 to 9: "tension". Above 9: "the reading ω_c = 3√3·ω_b does not read BAO + BBN".
- **Expiry.** The block expires on 31 December 2028.

Block A (secondary): a_0 with independent calibrations.

- **The target.** The first paper that measures g† of the radial acceleration relation, with the function of McGaugh and colleagues 2016. The total error must be $\leq 0.06 \cdot 10^{-10}$ m s^{-2}.
- **The rule.** z is computed against the $a_0 = cH_0/6$ of the chain. $|z| \leq 2$: "holds". From 2 to 3: "tension". Above 3: "does not hold".
- **Expiry.** The block expires on 31 December 2030.

Addendum 1. It was registered on 21 October 2026 at 00:29:34 UTC and corrected at 00:32:33 UTC. Both times before any DESI data beyond DR2, and with arXiv checked.

- The rule of block D does not change.
- Next to its outcome, the same calculation in the w_0w_a model is reported. A qualifier fixed in advance is added.
- The qualifier applies if three things happen together. Block D gives "tension" or worse. In w_0w_a $\Delta\chi^2_q \leq 4$. The same data prefer w_0w_a by at least 2σ ($\Delta\chi^2 \leq -6.18$). Then the text will say: "the outcome of block D depends on the background model".
- The qualifier is not given in one case: if 3√3 is saved only by a dark energy that already matters before the *drag epoch*. This happens when $w_0 + w_a > -0.3$. The drag epoch is the one in which ordinary matter stops being dragged by light: from then on the sound waves no longer propagate, and their distance stays imprinted. A test showed that this region can hide a wrong q. With synthetic data built with q = 6.5 one gets $\Delta\chi^2$ 91 in ΛCDM, but only 0.63 in w_0w_a.
- The program that decides the addendum passed 14 tests.

The program and its tests.

- A program decides the outcome. Its tests include a *dry run* on DR2, which reproduces the numbers above. They also include the counterexamples of the second verification round. They all pass.
- **A registered test was faulty.** It altered an element that was zero, so it never tested the asymmetry. 21 tests out of 22 had passed. It was redone with a non-zero element, and it passes.

7. What is known and what is specific to this analysis

What is already known. Many have already obtained H_0 from BAO + BBN, without the CMB:

- Addison and colleagues 2018 (66.98 ± 1.18);

- Cuceu and colleagues 2019 (67.6 ± 1.1);
- Schöneberg, Lesgourgues and Hooper 2019;
- Schöneberg and colleagues 2022 (67.6 ± 1.0; 68.3 ± 0.7 with ShapeFit);
- DESI DR2 (68.51 ± 0.58).

Krolewski, Percival and Woodfinden 2024 (arXiv:2403.19227) obtain H_0 from BBN by another route. They use a baryon fraction *measured* in the distribution of galaxies. Their error is wider (67.1 +6.3/−5.3). The H_0 of this work is the BAO + BBN value of the first group of analyses.

What is specific to this analysis.

- The ratio is imposed, q = Ω_c/Ω_b = 3√3, instead of being measured: one parameter fewer.
- With this come two more pieces. The first is the inverse reading, from the BAO to deuterium. The second is the link with $a_0 = cH_0/6$ and with the wide-binary chain.
- A targeted search found no analysis with this constraint together with BAO and BBN. It was not, however, a complete review.

What is not claimed.

- 3√3 is not derived. It is a reading chosen knowing the data (grade B*).
- With the CMB, 3√3 lies about 2σ from Planck (MM 2.44 and 2.54). That comparison is outside this chain.
- Verlinde's emergent gravity gives the same ratio if Ω_b = 4/81 (MM 2.54).

8. How this can turn out wrong

- **Block D gives $\Delta\chi^2 > 9$ with the DESI data beyond DR2.** Then the reading ω_c = 3√3·ω_b would not read BAO + BBN.
- **A precise measurement of g† far from $cH_0/6$** (block A).
- **A shift of the D/H average, or of the rates of the reactions d(p,γ)^{3}He and d+d, beyond the errors of D-1.** This would move ω_b, and so the whole chain. The D-5 dictionary shows how much it weighs to use the rates from before LUNA.

How the work was done

The main files.

- The data values are read on the sources. The SHA-256 of each file is in 00_fonti/SHA256_FONTI.txt.
- The Level 1 protocol is 02_livello1/PROTOCOLLO_LIVELLO1.md, registered on 20 October 2026 at 18:11:01 UTC.
- The code and the logs of Level 2 are in 03_livello2/.
- The check with the evolving dark energy is in the folder 07_energia_oscura. Its protocol quotes the author's request and is kept outside the archive (85e5a867…, 20 October 20:31:43 UTC). The outcome notes and the corrections after a verification round by Codex are in the folder. The supernova files are not redistributed.
- The prediction is 04_previsione/PREVISIONE_CATENA_DEUTERIO_v2.md. The arXiv check before registration is in 04_previsione/controllo_arxiv_v2. Addendum 1 is 04_previsione/AGGIUNTA_1_PREVISIONE_w0wa.md, with the program decidi_aggiunta_w0wa_v2.py.
- The program that decides the prediction is decidi_previsione_deuterio_v2.py. The faulty test that was redone is described in ESITO_PROVE_v2.md.

The registration. Every protocol and every program was registered with SHA-256 and UTC time before being run. The registers are the REGISTRO.txt files in each folder. The registered files were never modified: the corrections are separate files.

The Codex checks (GPT-6, model gpt-6-luna). Two rounds were done.

- **Round 1, on Level 1:** "not ready", 11 points. Ten were applied (02_livello1/CORREZIONI_LIVELLO1_1.md). One was not accepted, with the source: Y_p is compared in the PRyMordial convention Y_P = 4·Y_He (arXiv:2307.07061 eq. 2.20).
- **Round 2, on everything:** "not ready", 16 points, all applied (05_astra_giro2/CORREZIONI_GIRO2.md). They were mostly about the robustness of the decision program and the scope of T2a. T2a tests the expansion rate, not G separately.

- There was no third round on the scientific part.
- **A check of this package** (texts, run_all.py, build script): "not ready", 11 points, all applied. They concerned the wording of the public texts and a stricter run_all.py. No number changed (verifica_pacchetto/). No other round.

The files kept outside the archive. They are outside because they quote the author's words. Their SHA-256 hashes and registration times are in the registers, and they are available on request. The build script excludes them by name. Then it checks every packed file against a list of the author's phrases. The quotation marks in the correction files quote earlier texts by Claude, not by the author. They are:

- the premise;
- the Level 2 protocol (b3abd2a0…, 20 October 2026, 18:29:23 UTC);
- the Level 2 outcome notes (0d7506a9…, d1dbabe1…);
- the Level 0 inventory v1 (a40802fd…);
- the protocol of the w_0w_a check (85e5a867…);
- the messages sent to the verifier.

How to redo the calculations.

- **What you need:** Python ≥ 3.10 with numpy and scipy. For the full run, also classy (CLASS 3.4.1) and the PRyMordial code.
- `python scarica_dati.py` downloads the DESI DR2 likelihood files (commit b7b8a36e of CobayaSampler/bao_data) and PRyMordial (commit 725d8a8d). It checks their SHA-256: DESI 9ac154ab…, 252a1432…; BBN table bf4d1685….
- `python run_all.py` reruns the validations, Level 1, T1, the Level 2 comparison and the tests of the decision program. It compares every output with the reference logs. It takes about 12 minutes.
- A result is reported as DIFFERENT in three cases. The first: a program exits with an error. The second: a test is marked [NO]; the exception is the faulty registered test A12, redone separately. The third: a number differs by more than three units of its last printed digit.
- The two PRyMordial runs are compared on their JSON output with a relative tolerance of 0.5 %. PRyMordial's integration differs between Windows and Linux by up to 0.3 % in D/H and Li/H (verifica_pacchetto/confronto_piattaforme_bbn_linux.log). This is much less than the physical errors.
- The addendum tests (prove_aggiunta_w0wa_v2.py) are part of the normal run. The w_0w_a check with the supernovae (07_energia_oscura) needs the four supernova files. These files are not redistributed (SHA-256 of the pack in 00_fonti/SHA256_FONTI.txt). Without supernovae it works from the archive.
- `python run_all.py --completo` also reruns the PRyMordial runs, T2a, the inverse reading and the r_d grid. It takes about 25 minutes more, on a single core.
- The working texts (protocols, notes, corrections) are in Italian; the archived text summarises them.

Who did the calculations. The author is not a professional physicist. The calculations were done with AI assistants: Claude, with two checks by Codex. No human referee has examined them. Texts and materials: CC BY 4.0 licence. Code: MIT licence. Third-party data and code are not redistributed. The DESI files carry no explicit licence, and PRyMordial is GPL v3. scarica_dati.py downloads them.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **a_0.** The acceleration scale of galaxies. Here we read $a_0 = cH_0/6$.
- **BAO.** Baryon acoustic oscillations: a typical distance in the distribution of galaxies, used as a cosmic ruler.
- **BBN.** Big-bang nucleosynthesis: the formation of the first light nuclei in the first minutes.
- **χ^2 and $\Delta\chi^2$.** An error score and the difference between the scores of two models.
- **CMB.** The cosmic microwave background, the oldest light of the universe.
- **Deuterium.** Hydrogen with one more neutron in the nucleus.
- **D/H.** The number of deuterium nuclei for each hydrogen nucleus.
- **Dictionary.** Here: BBN used to translate laboratory data into cosmological quantities.
- **H_0.** The Hubble constant: how fast the universe is expanding today.
- **ΛCDM.** The standard model of cosmology.

- **Mirror sector.** A copy world of ours, with its own particles and its own light.
- **N_eff, ΔN_eff.** N_eff measures the very light and very fast particles of the young universe, besides photons, counted as families of neutrinos; ΔN_eff is how much more of them there is compared with what is expected.
- **Neutron lifetime (τ_n).** The typical time before a free neutron turns into other particles.
- **ω_b, ω_c.** How much ordinary matter and how much cold dark matter there are in the universe.
- **Prior.** A piece of information given to a calculation at the start.
- **q.** The ratio Ω_c/Ω_b between dark matter and ordinary matter.
- **σ.** The typical error.
- **Sound horizon (r_d).** The distance travelled by the sound waves in the young universe, up to the drag epoch.
- **w_0w_a (CPL).** A model of dark energy that changes over time.
- **Y_p.** The mass fraction of helium-4 produced in the first minutes.
- **z.** Redshift, or the z score (by how many σ a value is distant from another).

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